

Notes: Hart and Moore (QJE, 1994)

A Theory of Debt Based on the Inalienability of Human Capital

Contents

1	Big picture	3
2	Incremental contribution of the paper	3
3	Model setup	4
3.1	Players and project	4
3.2	Key assumptions	4
4	Debt contracts and two constraints	5
4.1	Promised repayments	5
4.2	Feasibility: repayments cannot be too early	5
4.3	Credibility: repayments cannot be too late	5
4.4	The central inequality	6
5	Debt capacity and project financing	6
5.1	Debt capacity	6
5.2	Proposition 1: when the project goes ahead	6
5.3	Corollary: when inefficiency disappears	7
6	Equilibrium repayment paths	7
6.1	Multiplicity	7
6.2	The slowest repayment path	7
6.3	The fastest repayment path	8
6.4	Proposition 2	8
7	Resolving repayment-path indeterminacy	8
7.1	Extension E1: debtor has reinvestment opportunities	8
7.2	Extension E2: creditor has better reinvestment opportunities	9
8	Comparative statics	9
8.1	More durable assets	9
8.2	More front-loaded project returns	9

8.3	More replaceable assets	10
8.4	Higher entrepreneur wealth	10
9	Figures	11
9.1	Figure I: model example	11
9.2	Figure II: renegotiation game	11
10	Connection to debt maturity in practice	12
11	Identification of the economic mechanism	12
12	Limitations and caveats	13
13	Short summary	13
14	Conclusion	14
14.1	Core conclusion	14
14.2	Economic intuition	14
14.3	Incremental contribution revisited	14
14.4	Implications	15
14.5	Takeaway	15

1 Big picture

- **Research question.** How can debt finance a profitable project when the entrepreneur cannot commit not to walk away with his human capital?
- **Core friction.** Physical assets are alienable: control of them can be transferred to the creditor. Human capital is inalienable: the entrepreneur can always quit or refuse to work.
- **Main idea.** The entrepreneur's ability to repudiate the contract limits how much future cash flow he can credibly pledge to the investor.
- **Debt capacity.** A profitable project may fail to be financed because the creditor cannot be promised enough future repayment without inducing future repudiation.
- **Equilibrium focus.** Because many contracts can implement the same allocation, the paper focuses on equilibrium repayment paths rather than on a unique legal contract.
- **Main result.** Feasible debt must balance two constraints: repayments cannot be too early, or the debtor runs out of cash; repayments cannot be too late, or the debtor repudiates and renegotiates.
- **Maturity result.** The model rationalizes the conventional rule that asset maturities and liability maturities should match: long-lived collateral supports longer debt, while front-loaded cash flows support shorter debt.
- **Economic takeaway.** Debt is not just a tax shield or a signaling device. It is a control-rights mechanism whose repayment schedule is shaped by collateral value, project cash-flow timing, and the borrower's ability to walk away.

2 Incremental contribution of the paper

- **Debt from incomplete contracting.** The paper gives a theory of debt based on the distinction between alienable physical assets and inalienable human capital.
- **Dynamic repayment paths.** Unlike static control-rights models, Hart and Moore study how repayment timing is constrained over the life of a project.
- **Endogenous debt capacity.** The maximum amount the entrepreneur can borrow is derived from the project's future cash flows and liquidation values.
- **Maturity structure.** The paper connects the maturity of optimal repayment paths to the maturity of project returns and collateral values.
- **Underinvestment mechanism.** Profitable projects may not be undertaken because the entrepreneur cannot pledge enough future surplus while retaining incentives not to repudiate.

- **Multiplicity of contracts, discipline of paths.** The deterministic model has many optimal contracts, including long-term contracts and sequences of short-term contracts, but the equilibrium repayment paths satisfy sharp bounds.
- **Link to practitioner wisdom.** The comparative statics match common lending rules: long-lived assets can support long-term borrowing, short-lived working-capital assets support short-term borrowing, and more entrepreneurial wealth makes lending easier.

3 Model setup

3.1 Players and project

There is an entrepreneur, denoted D , who has access to a profitable project but has insufficient initial wealth. The project costs K at date 0, while the entrepreneur has only $w_0 < K$. He therefore needs outside finance from a creditor, denoted C .

The project produces a deterministic flow of returns if the entrepreneur's human capital remains attached to the assets. Let

$$r(t) \geq 0, \quad 0 \leq t \leq T,$$

be the project return flow. The present value of continuing the project from date t is

$$R(t) = \int_t^T r(\tau) d\tau.$$

The physical assets also have an alternative use. If they are liquidated at date t , they generate a deterministic flow $l(\tau)$, so the liquidation value from date t is

$$L(t) = \int_t^T l(\tau) d\tau.$$

The rate of interest is normalized to zero.

3.2 Key assumptions

- **Positive surplus.** The project has positive net present value: $R(0) > K$. The initial liquidation value is not greater than the project cost: $K \geq L(0)$.
- **Entrepreneur-specific surplus.** Once the project is underway, continuation with the entrepreneur is efficient: $R(t) > L(t)$ for all relevant t .
- **No costless replacement.** The entrepreneur cannot be costlessly replaced. The surplus $R(t) - L(t)$ is the value of continuing with this entrepreneur rather than redeploying the assets.
- **Inalienable human capital.** The creditor can take control of physical assets after repudiation, but cannot force the entrepreneur to keep working.
- **No seizure of private savings.** The debtor's private savings cannot be seized by the creditor after repudiation. This assumption is important for the slowest repayment path.

4 Debt contracts and two constraints

4.1 Promised repayments

A contract specifies an initial investment I by the creditor and a stream of promised repayments $p(t)$ from the debtor to the creditor. The debtor's total outstanding indebtedness at date t is

$$P(t) = \int_t^T p(\tau) d\tau.$$

With competitive creditors and zero interest, the creditor breaks even in equilibrium, so $P(0) = I$.

4.2 Feasibility: repayments cannot be too early

The debtor must be able to make promised payments out of project returns, initial borrowing, and his own wealth. His private savings cannot become negative. This creates a lower bound on outstanding debt:

$$P(t) \geq \underline{P}(t) \equiv K - w_0 - R(0) + R(t).$$

Interpretation:

- If debt is repaid too quickly, $P(t)$ falls too fast.
- Then the entrepreneur has insufficient cash to keep meeting obligations.
- The fastest feasible path sets $P(t) = \underline{P}(t)$ at each date.

4.3 Credibility: repayments cannot be too late

At any date the entrepreneur can repudiate the contract. If he does, the creditor controls the physical assets and can either liquidate or renegotiate. The bargaining rule gives the creditor

$$\max \left\{ \frac{1}{2} R(t), L(t) \right\}$$

from date t onward after repudiation. Therefore, to make the original contract repudiation-proof, promised future debt cannot exceed this amount:

$$P(t) \leq \overline{P}(t) \equiv \max \left\{ \frac{1}{2} R(t), L(t) \right\}.$$

Interpretation:

- If future debt is too large, the entrepreneur can repudiate.
- Repudiation lets him renegotiate the outstanding debt down to what the creditor can secure through liquidation or bargaining.
- The slowest credible path sets $P(t) = \overline{P}(t)$ at each date.

4.4 The central inequality

Any feasible and credible repayment path must satisfy

$$\underline{P}(t) \leq P(t) \leq \overline{P}(t) \quad \text{for all } 0 \leq t \leq T.$$

This inequality is the core of the paper. It says that debt is possible only if the debtor can keep enough indebtedness outstanding to avoid cash shortfalls, while not keeping so much outstanding debt that repudiation becomes attractive.

5 Debt capacity and project financing

5.1 Debt capacity

Combining the feasibility and credibility constraints gives a maximum pledgeable amount, or **debt capacity**. Define

$$M = \min_{0 \leq t \leq T} \left\{ R(0) - R(t) + \max \left[\frac{1}{2}R(t), L(t) \right] \right\}.$$

The project can be financed if and only if

$$K - w_0 \leq M.$$

- $K - w_0$ is the amount the entrepreneur must borrow.
- M is the maximum amount the project can support as credible debt.
- If $K - w_0 > M$, the project is profitable but cannot be financed.

5.2 Proposition 1: when the project goes ahead

Proposition 1 states that the project goes ahead exactly when the required external finance does not exceed the debt capacity M . The paper also gives an equivalent expression:

$$M = \max \left\{ \frac{1}{2}R(0), \int_0^{t^*} r(\tau) d\tau + \int_{t^*}^T l(\tau) d\tau \right\},$$

where t^* is the first time at which either the creditor's bargaining claim is no longer dominated by liquidation, or project returns begin to dominate liquidation rentals.

Economic intuition:

- The creditor can always be promised at most either half of future project surplus or the liquidation value.
- If early liquidation rents exceed project cash flows, the debtor may need to hold part of the initial borrowing as a cash cushion to make early payments.

- This cash cushion reduces the amount of external funds available for the initial investment, lowering debt capacity.
- A larger entrepreneur equity contribution w_0 relaxes the financing constraint.

5.3 Corollary: when inefficiency disappears

If there is no initial sunk cost and project returns always dominate liquidation rentals, then the project is always undertaken. In particular, if $K = L(0)$ and $r(t) \geq l(t)$ for all t , the creditor can be protected by collateral while the debtor can pay out of project cash flows.

Economic message: underinvestment arises only when some combination of asset specificity, early low cash flows, or sunk costs prevents the entrepreneur from pledging enough future value.

6 Equilibrium repayment paths

6.1 Multiplicity

When $K - w_0 \leq M$, there is a continuum of optimal repayment paths. Any path satisfying

$$\underline{P}(t) \leq P(t) \leq \overline{P}(t)$$

is feasible, credible, and efficient. Along these paths the project is undertaken, the creditor breaks even, and liquidation does not occur on the equilibrium path.

The multiplicity arises because the no-repudiation constraint puts an upper bound on total future indebtedness at every date, but it does not pin down the exact timing of payments within that bound.

6.2 The slowest repayment path

The **slowest** path keeps outstanding debt as high as possible:

$$P(t) = \overline{P}(t) = \max \left\{ \frac{1}{2}R(t), L(t) \right\}.$$

Differentiating the debt stock gives the repayment flow. If liquidation value is the binding creditor outside option early on, then

$$p(t) = l(t).$$

Once the creditor's bargaining claim is binding, the repayment flow becomes

$$p(t) = \frac{1}{2}r(t).$$

Economic interpretation:

- The debtor borrows as much as the creditor can credibly be promised.
- Early repayments compensate the creditor for the depreciation of collateral.
- Later repayments share project cash flow.
- The debtor may need a cash cushion to cover early repayments when $l(t) > r(t)$.

6.3 The fastest repayment path

The **fastest** path keeps outstanding debt as low as possible:

$$P(t) = \underline{P}(t) = K - w_0 - R(0) + R(t).$$

This path has repayment flow

$$p(t) = r(t) \quad \text{for } 0 < t < T.$$

The debtor pays all project cash flows to the creditor until the final date. At date T , the creditor returns the residual surplus to the debtor:

$$P(T) = K - w_0 - R(0) < 0.$$

Economic interpretation:

- The debtor borrows only the minimum needed to start the project.
- The creditor receives all interim cash flows.
- The debtor effectively uses the creditor as a savings bank, receiving the residual at the end.

6.4 Proposition 2

Proposition 2 states that, conditional on the project being financed, the set of equilibrium repayment paths consists of all paths between the slowest and fastest bounds. The slowest and fastest paths are the two extreme cases, and later extensions select one of them as uniquely optimal.

7 Resolving repayment-path indeterminacy

7.1 Extension E1: debtor has reinvestment opportunities

If the debtor can reinvest project cash flows at a positive return while the creditor cannot, the slowest repayment path is uniquely optimal.

Reason:

- Delaying repayment lets the debtor retain more cash for profitable reinvestment.
- The creditor is still protected by the credibility constraint.
- Thus the debtor wants the maximum credible indebtedness at each date.

Economic message: when the entrepreneur has high-return internal uses of cash, optimal debt is back-loaded as much as credibility allows.

7.2 Extension E2: creditor has better reinvestment opportunities

If the creditor can reinvest cash more productively than the debtor, the fastest repayment path is uniquely optimal.

Reason:

- It is inefficient for the debtor to hold idle or low-return cash.
- The debtor should transfer project cash flows to the creditor as early as feasible.
- The fastest path implements this allocation.

Economic message: when creditor-side reinvestment opportunities dominate, optimal debt is front-loaded.

8 Comparative statics

8.1 More durable assets

If assets become more durable, then $L(t)$ rises for future dates. Proposition 3A implies:

- **Debt capacity rises.** The project is more likely to be financed.
- **Slowest debt becomes slower.** Better collateral lets the creditor wait longer before being repaid.
- **Fastest debt is unchanged.** The fastest path depends mainly on the project return stream and the amount initially borrowed.

Economic message: durable collateral supports long-term debt.

8.2 More front-loaded project returns

If project returns become more front-loaded while total project value is held fixed, Proposition 3B implies:

- **Debt capacity rises.** Early cash flows make it easier for the debtor to meet early obligations.

- **Slowest debt becomes faster.** The creditor has less future surplus to bargain over later.
- **Fastest debt becomes faster.** More of the cash arrives early and can be paid out early.

Economic message: projects with early cash inflows can support shorter-maturity debt.

8.3 More replaceable assets

The paper models greater asset replaceability as a reduction in the creditor's bargaining power after repudiation. Proposition 3C implies:

- **Debt capacity falls.** If assets are easier for the debtor to replace, the creditor has less leverage.
- **Slowest debt becomes faster.** Weaker creditor control requires earlier repayment.
- **Fastest debt is unchanged.**

Economic message: assets that are highly replaceable from the entrepreneur's perspective provide weaker discipline and support less long-term debt.

8.4 Higher entrepreneur wealth

If the entrepreneur's initial wealth w_0 rises, the project is more likely to go ahead because less outside finance is needed.

In the fixed-scale model:

- the slowest path is unaffected;
- the fastest path becomes faster because the initial loan is smaller.

In the variable-scale extension, higher initial wealth increases project scale and increases both the amount initially borrowed and the repayment path proportionately.

9 Figures

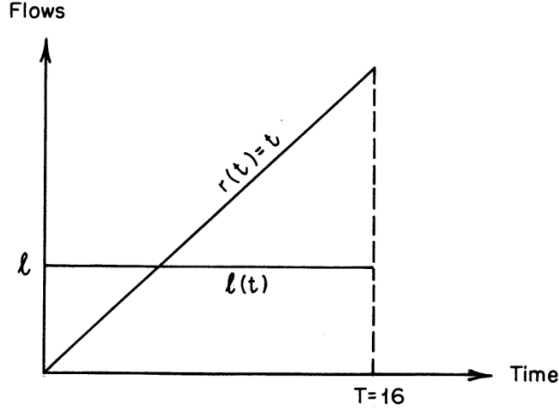


FIGURE I

Figure I: Example with increasing project returns and a constant liquidation flow.

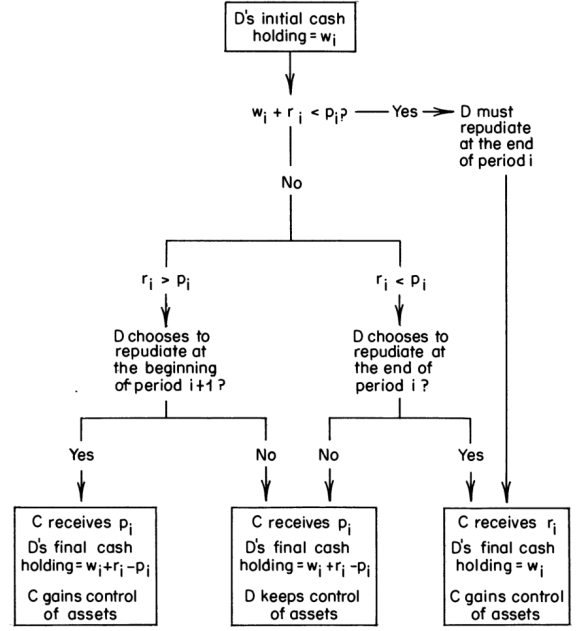


FIGURE II

Figure II: Discrete-time renegotiation game in Appendix A.

9.1 Figure I: model example

Read:

- Figure I illustrates an example with horizon $T = 16$, increasing project return flow $r(t) = t$, and constant liquidation flow $l(t) = \ell$.
- The upward-sloping line is the return flow generated when the entrepreneur's human capital remains with the project.
- The horizontal line is the liquidation or alternative-use flow of the physical assets.
- Early in the project, liquidation may generate more immediate flow than continuation; later, the entrepreneur-specific project dominates.

Economic message: the figure shows why early cash-flow shortages can bind even when the project is profitable in present value. If liquidation flow exceeds project cash flow early on, the debtor may need a cash cushion to make early repayments credible to the creditor.

9.2 Figure II: renegotiation game

Read:

- Figure II appears in Appendix A and summarizes the discrete-time renegotiation logic.
- The key state variable is the debtor's cash holding before a payment is due.
- If the debtor cannot pay, he must repudiate and the creditor gains control of the assets.
- If the debtor can pay, he may still choose to repudiate if doing so improves his continuation payoff.
- The equilibrium payoff to the creditor after repudiation is governed by the larger of liquidation value and the creditor's bargaining share of continuation value.

Economic message: the figure clarifies why debt contracts must be both feasible and credible. The debtor must have enough cash to pay, and the outstanding debt must not be so high that repudiation and renegotiation become attractive.

10 Connection to debt maturity in practice

The discussion section connects the theory to conventional lending practice.

- **Good projects borrow more easily.** Higher total returns $R(0)$ increase debt capacity.
- **Collateral matters.** Higher liquidation value or more durable collateral increases the creditor's security and supports more borrowing.
- **Entrepreneur equity matters.** Larger w_0 reduces the required external finance and makes the project more likely to go ahead.
- **Long-lived assets support long-term debt.** Real estate and durable fixed assets can credibly back longer repayment schedules.
- **Working capital supports short-term debt.** Inventory, payroll needs, seasonal financing, and accounts receivable are associated with faster cash-flow realization and shorter debt.
- **Matching principle.** The model formalizes the rule that assets should be matched with liabilities: debt repayment should track either the project return stream or the depreciation of collateral value.

11 Identification of the economic mechanism

This is a theory paper, not an empirical paper, so “identification” means isolating the mechanism that generates debt and maturity predictions.

- **What drives debt capacity.** The debt capacity M is determined by the interaction of cash-flow timing $r(t)$, liquidation value $L(t)$, and the debtor's ability to repudiate.
- **Why collateral matters.** Collateral gives the creditor an outside option after repudiation, increasing the set of credible repayments.

- **Why maturity matters.** Back-loaded repayments preserve debtor liquidity but weaken creditor protection; front-loaded repayments protect the creditor but can exhaust debtor cash.
- **Why human capital matters.** If the entrepreneur could be forced to continue working or could be costlessly replaced, the creditor could capture more surplus and the debt-capacity constraint would be much weaker.
- **Why renegotiation matters.** Even after repudiation, the parties may renegotiate because continuation is efficient, but renegotiation does not eliminate inefficiency because the debtor still cannot pledge all future cash flows.

12 Limitations and caveats

- **Deterministic environment.** The main model assumes certain project returns and liquidation values. Uncertainty would make the analysis more realistic and could pin down contract form more sharply.
- **Single creditor.** The baseline model has one creditor. Multiple creditors could make renegotiation harder, which may strengthen commitment but also increase inefficient liquidation risk.
- **Specific bargaining rule.** The paper uses a simple 50:50 surplus split with a liquidation outside option. Different bargaining protocols may change quantitative results but preserve the core pledgeability logic.
- **No explicit equity security.** The model discusses equity-like initial wealth and control rights, but it does not fully model outside equity finance.
- **No consumption during the project.** The baseline model assumes the debtor consumes only after the project. Allowing interim consumption could change the selected repayment path.
- **Physical asset indivisibility.** Liquidation is modeled as transfer of control over the asset as a whole, rather than partial liquidation.

13 Short summary

- The paper develops a theory of debt based on the inalienability of human capital.
- An entrepreneur needs funding for a profitable project but can always withdraw his human capital.
- Because the creditor can seize physical assets but not the entrepreneur's labor, future repayments are only partly pledgeable.

- A repayment path must be feasible, meaning the debtor can afford to pay, and credible, meaning the debtor does not want to repudiate.
- These two constraints generate a debt-capacity condition: the project is financed only if required borrowing does not exceed M .
- If financed, there is a continuum of efficient repayment paths between a fastest and a slowest path.
- Debtor reinvestment opportunities select the slowest path; creditor reinvestment opportunities select the fastest path.
- More durable assets and more front-loaded project returns increase debt capacity.
- Durable collateral supports longer-term debt, while front-loaded cash flows support shorter-term debt.
- The model provides a theoretical foundation for the matching of asset and liability maturities.

14 Conclusion

14.1 Core conclusion

Hart and Moore show that debt capacity and debt maturity can be understood as consequences of incomplete contracting. Since the entrepreneur's human capital is inalienable, the creditor cannot force him to generate future returns. This limits the amount of future cash flow the entrepreneur can pledge and can prevent profitable projects from being financed.

14.2 Economic intuition

Debt works because control over physical assets gives the creditor bargaining power after default. But this power is limited. If repayments are due too quickly, the debtor runs out of cash. If they are postponed too far into the future, the debtor may repudiate and renegotiate. Optimal repayment paths lie between these two forces.

14.3 Incremental contribution revisited

The paper's main contribution is to turn a qualitative idea - "debt is backed by assets" - into a dynamic theory of debt capacity and repayment timing. It shows exactly how project returns, liquidation values, and entrepreneurial wealth determine whether debt can finance a project and how fast repayment can occur.

14.4 Implications

The theory explains why durable fixed assets support long-term debt, why working capital is financed with short-term debt, and why borrower equity matters. It also suggests why asset specificity and poor collateral value push firms away from conventional debt and toward other governance or financing arrangements.

14.5 Takeaway

The enduring takeaway is that debt is constrained by what the borrower can credibly promise without surrendering his human capital. The maturity of debt should match the timing of project cash flows and collateral value because both determine the creditor's control and the debtor's incentives to repay.